Pupil booklet

A5 printed

ENGINEERING SCIENCE

Revised June 2022

Energy, System and Control

Topic 1: Types of energy and the transfer of it.

Topic 2: The universal system diagram (block diagram).

Topic 3: Sub-systems diagrams.

Topic 4: Open/Closed loop systems and error detection.



Name: _____

Class: _____

Teacher: _____

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Outcomes and Assessment Standards

There are main 3 outcomes in this unit and it has been divided into three booklets. By the end of this second booklet, you will have completed outcomes 1 and 3. You will investigate engineering systems, problems and solutions, involving some existing and emerging technologies and consider implications relating to the environment, sustainable developments, and to economic and social issues.

The Context and Challenge unit Outcomes are:

1. Investigate engineered objects.
2. Investigate engineered challenges and relate these too key engineered concepts.
3. Describe some aspects of the impact of engineering.

Outcome 1

The candidate will:

1 Investigate engineered objects by:

1.1 Describing, using the systems approach, how some engineered objects work.
1.2 Identifying sub-systems and describing the function of each and how they interact.
1.3 Producing system diagrams to show sub-systems.
1.4 Carrying out energy audits.

Outcome 2

The candidate will:

2 Investigate engineering challenges and relate them to key engineering concepts by:

2.1 Identifying and describing how several different branches of engineering contribute to solving an engineering challenge.
2.2 Describing examples of the varied roles of engineers in designing, implementing, testing and controlling complex systems.
2.3 Modelling some aspect (related to one branch of engineering) of a solution to an engineering challenge.
2.4 Explaining how emerging technologies may provide improved solutions to engineering challenges.

Outcome 3

The candidate will:

3 Describe some aspects of the impact of engineering by:

3.1 Describing examples of social and economic impacts of engineering.
3.2 Describing some examples of environmental impacts of engineering.
3.3 Describing some ways in which engineering solutions contribute to tackling climate change.

Outcomes and Assessment Standards: System and Control

Learning Intentions

- To learn about the different roles of engineering.
- To learn about the varied role of Engineers in designing, implementing, testing and controlling complex systems.
- To learn the effects that engineering has on the world.
- To know about control diagrams for simple control systems.
- To learn about what the systems approach is.
- To learn about subsystems.
- To learn about energy forms.
- To learn about basic energy transformations.
- To know about open loop & closed loop control.
- To know what a feedback loop is.
- To learn what an error detector is and how it's used in a control system.

Success Criteria

Use traffic lighting to record your progress. This will also help with revision as you will be able to see what areas you need to work on. Your teacher will sign off this page after speaking to you about your self-evaluation:

Red: I need to work on this topic further to understand it.

Amber: I need a bit of help to fully understand this topic.

Green: I am confident in my understanding of this topic.

- ☐ I can identify and describe different branches of engineering.
- ☐ I can explain how emerging technologies can provide improves solutions to engineering challenges.
- ☐ I can identify basic energy transformations.
- ☐ I can describe examples of social, environmental and economic impacts of engineering.
- ☐ I can draw and understand a system diagram, showing inputs and outputs.
- ☐ I can describe, using the systems approach, how some simple engineered objects work.
- ☐ I can identify the main sub-systems of an object.
- ☐ I can draw and understand sub-system diagrams.
- ☐ I can describe the operation of systems from reading their system/sub-system diagram.
- ☐ I can understand and create diagrams for simple control systems.
- ☐ I can understand the difference between automatic and manual control.
- ☐ I can understand how an error detector is used within control systems.

What is energy?

Energy is the ability to do work. The 'work' comes in many different forms – from a car driving along, to a television set showing a programme, to a human body running a race. In order to make things or machines happen/move, they need energy. Energy is all around us and comes in many different forms.

Once harnessed or created, energy cannot be destroyed, it can only be transformed, from one type into another. Indeed, this is what all machines and living things do – transform energy. For example, a car converts or transforms the chemical energy in petrol into heat energy inside the engine. The engine then turns the heat energy (combustion of the fuel) into mechanical energy, which makes the car move. A television set converts or transforms electrical energy into light energy - to give the picture - and into sound energy for the soundtrack. Humans get the energy they need for their busy lives from the chemical energy in food. They transform it into the energy of movement and heat energy.

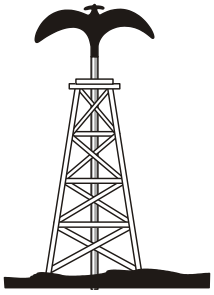
Where does energy come from?

The Sun is the ultimate, direct and indirect source of almost all of our energy. This is most obvious in the form of the light and heat it provides but it has an impact on our weather systems (wind and rain) and provides the energy plants need to grow, providing us with energy in the form of the food we eat. Even fossil fuels like coal and oil required the sun to grow the raw ingredients that formed them. Without the Sun, nothing would exist. Energy can also be created in one of three ways:

Nuclear *Renewable* *Biosphere/finite fuels*

The 'biosphere' is made up of the parts of Earth where life exists. It extends from the deepest root systems of trees, to the dark environment of ocean trenches, to lush rain forests and high mountaintops.

Oil and Gas energy



A non-renewable resource is a resource of economic value that cannot be readily replaced by natural means on a level equal to its consumption. Most fossil fuels, such as oil, natural gas and coal are considered non-renewable resources (finite) in that their use is not sustainable because their formation takes billions of years. Animals, as well as plants, died and settled on the seabed and became buried. Under immense Earth pressure and heat, decaying material changed into oil. Some of the oil decayed even more and made natural gas. These oil and gas reserves supply the World with precious resources that are used today at a faster rate than new reserves being naturally created.

Before it can be used, natural gas is purified. The methane is then ready to be piped directly to homes and factories. Oil that comes directly from wells is called crude oil. This is because it is not just one substance but many, all mixed together. It has to be treated before it can be used and the factory where the oil is treated is called a refinery. Crude oil is refined to separate it into products like Liquid Petroleum Gas, Petrol, aviation fuel, pharmaceuticals, chemicals from which to make plastics and many other products.

Nuclear energy

Radioactive uranium ore is used to generate nuclear power. It is mined from the ground in countries all over the world. Compared with other fuels, it is a very concentrated energy source, which means that a great deal of energy can be released from quite a small amount.

The energy in uranium cannot be released by burning. Uranium must be split into smaller atoms to obtain the energy. This decay process can happen naturally in the ground, but in a nuclear power station it is speeded up in a machine called a reactor. This process is called nuclear fission and can be carefully controlled in the reactor to produce a steady supply of heat. This heat energy is used to produce steam, which turns turbines.

Although it seems to be a very efficient way of producing energy, there are concerns about the safety of these reactors and the large amounts of dangerous radioactive waste that they produce. One way to dispose of this waste is to bury it in thick concrete and lead lined container bunkers buried deep underground. The fear however is that the containers could crack and release the radioactivity sometime in the future. It is important to note that the waste material remains highly radioactive for thousands of years.

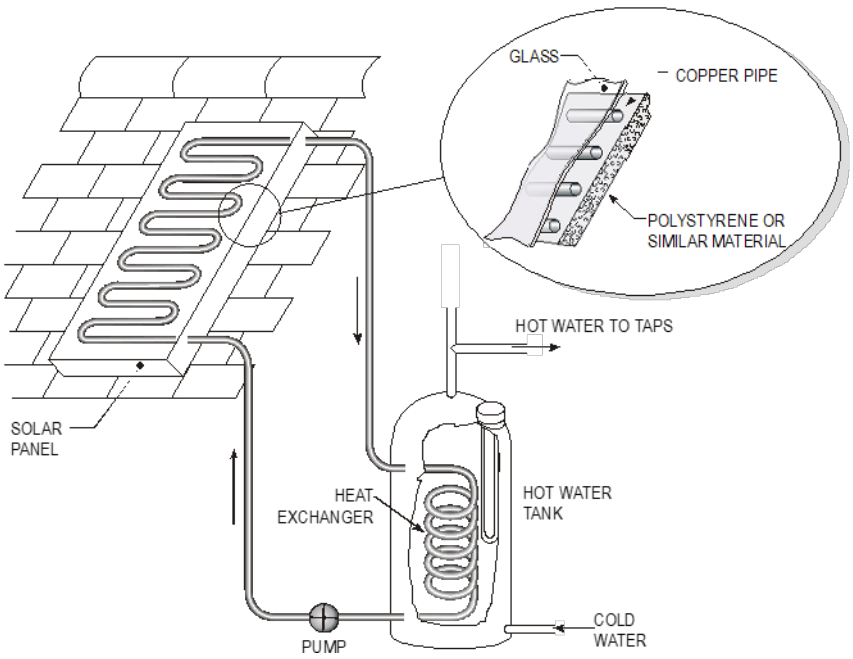
Renewable energy sources

Today, we use so much energy that we depend very heavily on the fossil fuels that have taken nature billions of years to make. There will be a time when the World's limited resources are used up. Renewable energy is defined by an energy from a source that is not depleted when used, such as wind, solar power, hydroelectricity and energy from the sea.

Every day, the Earth absorbs vast quantities of the Sun's energy. Even if only a small proportion of this could be harnessed, there would be enough energy for our needs over and over again, just as long as the Sun continues to shine and the Earth is still alive. Energy sources that come from daily sunshine can be harnessed domestically on a small scale (house roof/shed), but also on a larger industrial scale (on the side of tall buildings) if the location ensures long periods of clear, dry and direct sunshine. The location is also important to ensure no shadows from other objects that may block the Sun's rays. Large areas of countryside land are also being converted to Solar energy capture. Areas of America has significant solar energy capture and has done so for many years. Efficiency in energy capture is improving through the continual development in technology.

Solar energy

Science and technology now allow us to trap the enormous power of the sun and use it to produce heat and electricity. The word Photosynthesis comes from plants that absorb sunlight and turn that energy into food. This is a compound word made up of photo (which means "light") and synthesis (which means "to put together"). Buildings can be designed to absorb heat on 'heat collectors' during the day and release it when it is most needed, at night. Solar panels on rooftops work in the same way, but the heat is stored in a liquid such as water or oil. This is used to heat the building by passing it around a central heating system, or simply to provide hot water.



The Sun's light energy can be turned into electricity using photovoltaic cells that contain crystals of silicon. These cells work in a similar way to a car battery, but instead of being recharged by something that moves, such as an engine, they are recharged by light. They were first made for spacecraft, but now you may find them powering your calculator or watch.

Solar heat can be captured on a vast scale to generate electricity. The sun's light is reflected off many mirrors and on to one small target, to concentrate the energy. This means that much more of the sun's radiation is collected in one place. The intense heat can then turn water into steam to turn a turbine.

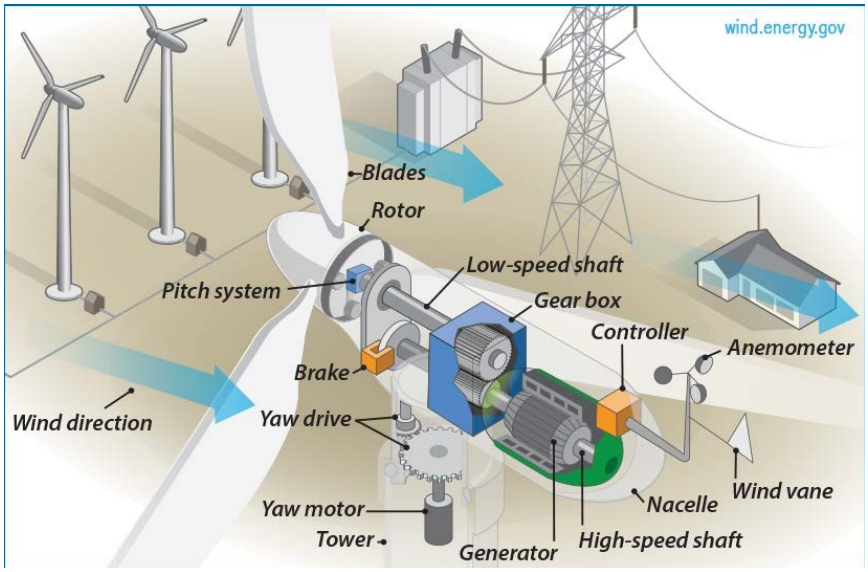
Wind energy



In the Netherlands, wind power has been used to turn windmills for thousands of years. They were used to turn machinery to raise water from deep wells and to grind wheat. In recent years, new ways have been found to capture the wind's energy more efficiently. However, because winds vary from place to place, on land or out in sea and from day to day, this can prove to be difficult. Wind energy turbines does cause significant damage to the local geography and many are seen as an eyesore.

So how do wind turbines make electricity? Simply stated, a wind turbine works the opposite of a fan. Instead of using electricity to make wind, like a fan, wind turbines use wind to make electricity. The wind turns two or three propellered-like blades round a rotor, which spin a shaft, which connects to a generator and makes electricity.

Wind is a form of solar energy and is a result of the uneven heating of the atmosphere by the Sun, the irregularities of the earth's surface, and the rotation of the Earth. Wind flow patterns and speeds vary greatly across the country and are modified by bodies of water, vegetation and differences in terrain. Humans use this wind flow or motion energy, for many purposes: sailing, flying a kite, and even generating electricity.



Effects of wind turbines

Although wind power plants have relatively little impact on the environment compared to fossil fuel power plants, concerns have been raised over the noise produced by the rotor blades, visual impacts, and deaths of birds and bats that fly into the rotors (bat mortality)

Noise

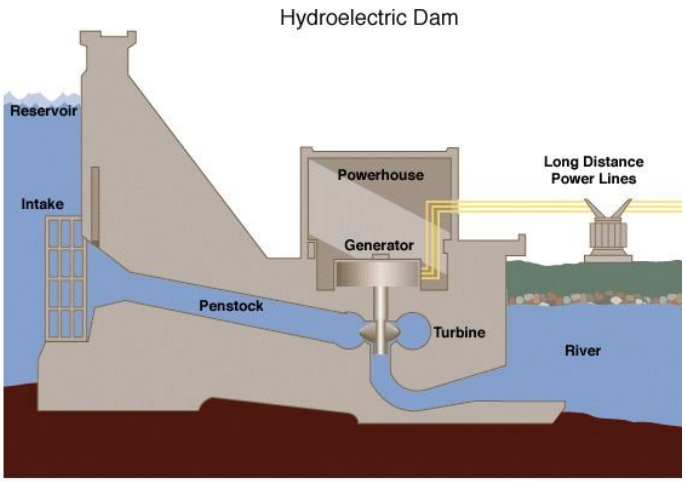
Like all mechanical systems, wind turbines produce some noise when they operate. Most of the turbine noise is masked by the sound of the wind itself, and the turbines run only when the wind blows. In recent years, engineers have made design changes to reduce the noise from wind turbines. Early model turbines are generally noisier than most new and larger models

Visual impacts

Because they must generally be sited in exposed places, wind turbines are often highly visible; however, being visible is not necessarily the same as being intrusive.

The land use impact of wind power facilities varies substantially depending on the site: wind turbines placed in flat areas typically use more land than those located in hilly areas. However, wind turbines do not occupy all of this land; they must be spaced approximately 5 to 10 rotor diameters apart (a rotor diameter is the diameter of the wind turbine blades). Thus, the turbines themselves and the surrounding infrastructure (including roads and transmission lines) occupy a small portion of the total area of a wind facility. The amount of physical engineering (civil, mechanical, electrical, environmental) required for the installation, maintenance and the carbon footprint of manufacturing the turbines themselves are all things which need to be carefully balanced with the effects on the environment. Offshore wind farms have much larger rotor diameters to harness stronger winds and some people seen such schemes has having less of a visual impact due to the distance away from the land.

Hydroelectric energy



Water power has been used for centuries. Nowadays, hydroelectric schemes are used to generate large amounts of electricity. Usually a dam is built across a river valley to control the flow of water and store up the energy. The water collects in a reservoir behind the dam. When electricity is needed, the water is allowed to rush through holes in the dam, turning turbines as it flows. Once built, it is an economic system as the water which passes can either be pumped back up when energy cost is low or it supplies a dam further down the water course. The disadvantages are the expense and civil engineering involved to create the dam, limited to the correct landscapes and long periods of hot temperature may also reduce the amount of water available (drought).

Energy from the sea

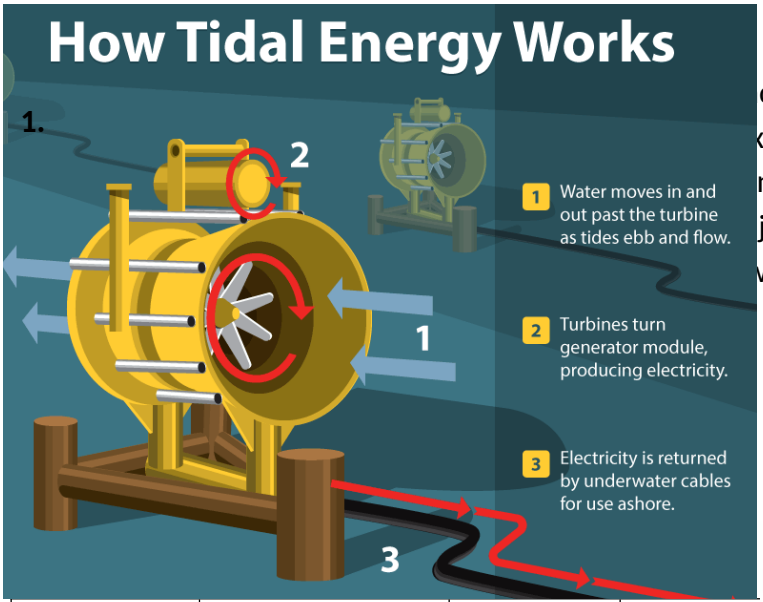
The sea can provide us with energy for generating electricity in several ways. Some of these ideas are still being tested and improved. By placing energy systems out at sea, many people believe they cause less of an eyesore. The power however still needs to reach land by HV cables and these areas of developments does cause a hazard for ships and fishing nets due to the restrictions around them but also marine life.

Wave energy

Waves happen because winds transfer their energy to the sea's surface as they blow over the water. Wave power systems use kinetic energy in the waves to turn turbines. These can be either floating or fixed to the seabed, and placed either out at sea or on the shoreline.

Tidal energy

Tides are caused mainly by forces from the Moon and the Sun. This force is called gravity and it pulls on the water. Just like wind turbines, they can be placed on the sea bed and as the water moves in and out with the tide, the blades will turn to generate electricity. Tidal energy can also be harnessed by building a barrier right across a river estuary. Electricity is generated in the same way as with a hydroelectric dam, except that it is the push and pull of the tides that causes water to flow over the turbines. However, people fear that tidal turbine schemes may upset estuary environments that are rich in wildlife.



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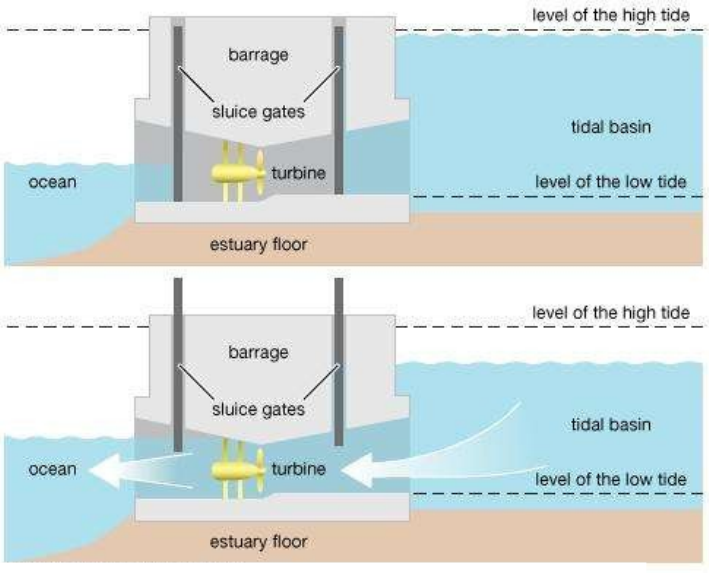


Above is a CAD model of a sea floor turbine.

Source of fuel	Type of fuel:	Advantages?	Disadvanta



Above is a photograph of a tidal turbine.



Concept diagram of a barrier across an estuary.

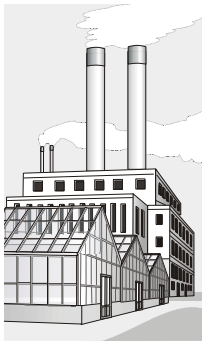
Gas	Finite (fossil fuel)	Easy to store, reliable	Non-renewable. Cost of extraction very high. Highly explosive.	Colourless and odourless when extracted but added during Chemical processing. Burns blue/orange flame.	Gas fields found in the deep North sea.	Domestic and industrial for heating and hot water. Used to generate electricity (power stations)
Coal						
Oil						
Wind						Investigate the 'Seagreen' development
Sun						
Solar panels						
Geothermal						
Biomass						
Hydro-electric						
Tidal						
Nuclear						
Hybrid						

Energy and the Environment

The way energy is produced can have a huge impact on the environment so one key role for engineers is to ensure it is produced and used in ways that minimise negative impacts.

Climate change - The 'Greenhouse effect' and the carbon footprint

At present, fossil fuels supply most of the world's power, but when any of these fossil fuels are burnt, carbon dioxide gas is released. There is a small amount of carbon dioxide naturally present in the air, but because fossil fuels are used so much, it is building up in the atmosphere. In these unnatural quantities, carbon dioxide is one of the gases that cause the greenhouse effect.



Gases such as carbon dioxide allow the Sun's heat to enter the atmosphere but stop it escaping back into space. Heat is trapped in a layer around the Earth. As a result of the increased amount of carbon dioxide, temperatures have risen around the world. The air inside greenhouses heats up in a similar way. The glass lets in the sun's heat but stops it escaping back out again. In this way the glass is like the layer of carbon dioxide. Because of this similarity, global warming has become known as the 'greenhouse effect'.

You might think that it would be very pleasant to have a warmer climate, but some scientists think the effect could be very serious. Some animals and plants are very sensitive to temperature, and so **global warming** could cause them to die out. Also, as the weather gets warmer, the ice caps at the North and South poles will gradually melt. The melting of the southern ice cap will cause sea levels to rise. This will lead to flooding, reducing areas of land and killing off even more living things. Changes in global weather patterns can also be attributed to Global warning.

Acid rain

As rainwater falls through the atmosphere, it mixes with carbon dioxide gas and becomes an acid called carbonic acid. This natural acid is so weak that it does not harm the environment.

However, when coal is burned, sulphur and nitrogen in the coal combine with oxygen to give oxides. These mix with rainwater and make it much more acidic. The main acids that this process produces are sulphuric and nitric acids, which are very strong. This acid rain is polluting the environment, causing trees to stop growing and animals and plants in lakes and rivers to die. It also corrodes buildings, making them crumble.

It has become vitally important for scientists and technologists to solve the problem of producing or harnessing energy from the Earth without having to burn fossil fuels.

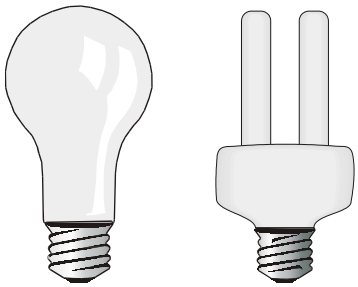
Oil pollution

Oil pollution can cause huge problems. Leakage and spills from oil tankers can release tonnes of oil into the environment. All sorts of marine life and birds are destroyed, and beaches are covered in oil. The pollution is very difficult and expensive to clear up properly and safely. The problem is made worse by high winds and rough seas, which break up and spread oil slicks that are already many square kilometres in size.



Energy efficiency in the home

Many homes used to have incandescent light bulbs like the one on the left in the diagram below. Many homes used to have traditional incandescent light bulbs like the one on the left of the diagram below but they are very inefficient – they waste 95% of the electricity they use by turning it into heat! In 2009, under new EU rules, these bulbs were phased out to be replaced by energy efficient light bulbs – a measure that will reduce co2 emissions by up to 15 million tons a year. They can also last anywhere between twenty-thirty times longer (1000 hours compared against 30 000 LED).



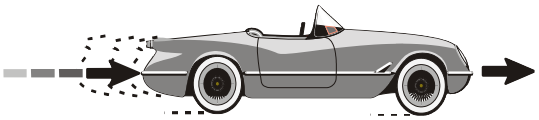
Forms of energy

Energy is defined as the capacity of a body to make things happen (or to work). **It is measured in joules (J)**. We know that to do more work we require more energy, but we can use various types of energy to do this work for us. Energy comes in several different forms, but the ones we shall deal with in this course are:

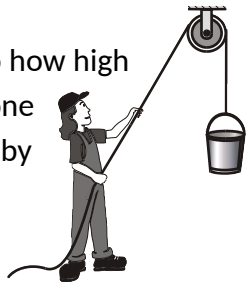
- kinetic (movement)
- potential (stored)
- magnetic (field)
- Electrical
- heat
- light
- sound

Kinetic energy

Kinetic energy is the energy of movement. It is the name given to the energy a body possesses due to its motion. The car in the diagram below can be described as having kinetic energy because it is moving.



Potential energy



Potential energy can be best thought of as energy **stored** in a static object. It can be due to how high the object is above a datum (starting point), or due to the fact that work has already been done on the object and the energy is stored in it (for example in a spring). The bucket supported by the pulley has potential energy until it is released.

Magnetic field energy

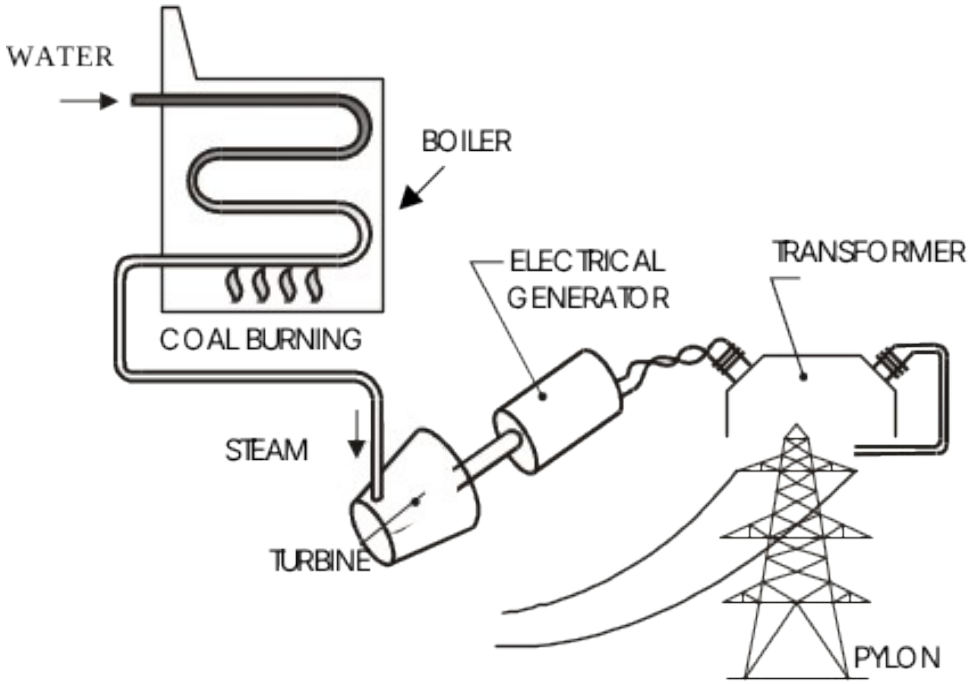
A magnet is an object that is made of materials that create a *magnetic field*. Magnets have at least one North pole and one South pole. A magnetic field is the region in space where a magnetic force can be detected. Magnetism is the force of attraction or repulsion between substances made of certain materials, such as iron, nickel, cobalt, and steel. The force of magnetism, simply put, is due to the motion of electric charges.

Magnets are present in most electronic devices. In fact, anything that has a motor uses a magnet. Televisions, computers hard drives and microwave ovens all operate with magnets. Magnets are used to keep refrigerator doors closed and new electric train use magnets to lift them off the ground so that they float. Floating reduces friction and allows the train to run more efficiently. Magnets are used to slow down roller coasters. You'll also find magnets in medical devices. Magnetic Resonance Imaging (MRI) is a type of scan that uses strong magnetic fields and radio waves to produce detailed images of the inside of the body. More uses for magnets are found every day.

Electrical energy

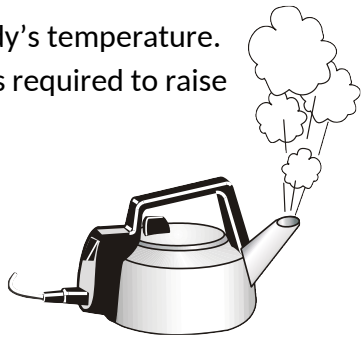
Electrical energy is one of the most convenient and commonly used forms of energy since it can be transported easily from place to place (along electrical cables) and can be easily changed into other forms of energy. Most electrical energy is generated in power stations, where one type of energy is converted into electrical energy.

The diagram below shows a simple fossil fuel power station which is based on Scottish and Southern Energy's Power Station in Peterhead. The fuel, which is various gases brought from the North Sea, is burnt in a boiler and the chemical energy in the fuel is converted into heat energy. This heat energy is used to produce steam at very high pressure (kinetic energy). The high-pressure steam is then used to turn turbine blades (rotational kinetic energy). The turbine is in turn connected to an alternator, which converts the kinetic energy into electricity. This electricity can now be transported around the country along electrical pylons and cables.



Heat energy

Heat energy is the energy transferred to a body that results in a change in the body's temperature. The diagram below shows a kettle boiling: a certain amount of thermal energy was required to raise the temperature of the water in the kettle to boiling point



The system approach

Now we know about the background of Engineering and how it leads to advancements in technology. In order to solve an everyday problem or improve existing technology, an analysis of all the inputs and outputs involved in the system must first be carried out. Solving problems in technology can be made simpler if we use something called the SYSTEMS APPROACH and consists of these three basic elements.

INPUT PROCESS OUTPUT
(control)

The 'arrows' represent the direction of energy flow. Most systems have more than one input and output.

For most things we want it to do something for us. This is called the OUTPUT.
To get it to do this we need to put something in. This is called the INPUT.

How this is done is called the PROCESS. We do not need to know how it is done so the process is usually put inside a 'Black Box.' This block diagram is called the UNIVERSAL SYSTEM DIAGRAM and can be applied to any system.



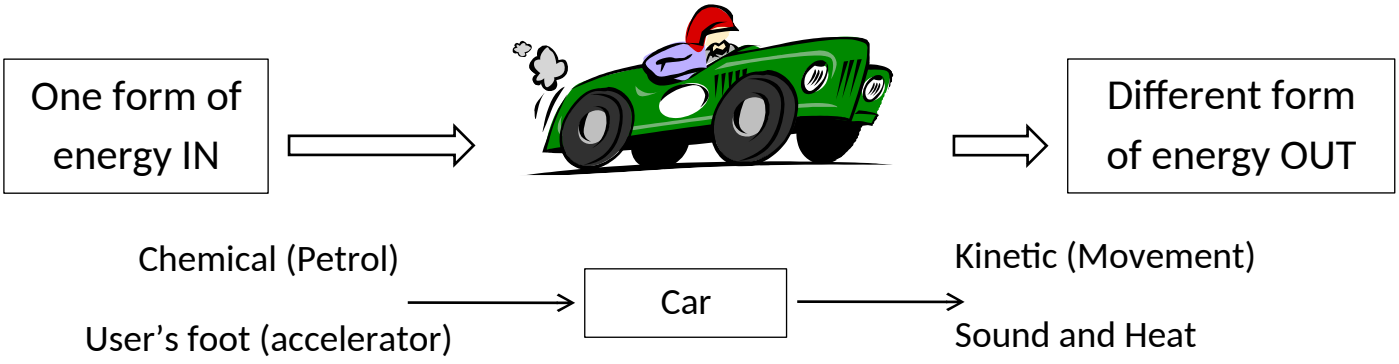
2. What is Newton's third law of motion? Research it and write the answer in your jotter.

It might help you to understand the idea of a system if you think about some of the systems that are at work in your own body. For example: if you pick up a hot plate by mistake, you quickly put it down. Your fingers sense the heat and pass a signal to your brain. Your brain processes (decides what to do) this signal and sends its own signal to the muscles in your arm, then your muscles pull your hands away from the plate. When going to work in the morning a person can use a car to get there. They only needs to know how to drive and to put petrol into it. How it all works can be left to the mechanics in the garage.

Energy is needed to make things work

ENERGY can be CONVERTED from one form into another by a suitable process. Energy CANNOT just disappear. It might reduce but it must be transformed into something else.

Take our example of the car from before its input is petrol which is a type of CHEMICAL ENERGY. Its output



Of course, there are also many other inputs and outputs to a car.

Main forms of energy

The purpose of some systems is to convert energy from one form to another. This means that we also need to consider all the energy inputs and outputs going into and out from a system.

3. In your jotter, write a brief statement in your own words which explains each '*form*' of energy below.

Sound	*Heat*	*Light*
Electrical (not electricity)	*Chemical*	*Nuclear*
Magnetic (field)	*Kinetic (movement)*	*Potential (stored)*

4. Create a block diagram (universal system diagram) for each system below by adding the correct inputs and outputs. State the action and not the energy. Answer in jotter.

(input) W*hat has to happen for it to work?*
(output) *What does it do?*

- Doorbell
- Flour windmill
- Table Lamp
- CD Player
- TV
- Hair dryer

5. Create a block diagram (universal system diagram) for each of the following systems— showing all energy inputs and outputs. Answer in jotter.

- Kettle
- Coal fire
- Corded drill
- Radio
- Bike dynamo (lights)
- Hydro power station

Energy transfer

Energy transfers between different types in almost every system in the world, natural or manmade. It is important that engineers can identify the transfer of energies within a system. Usually these transfers are quite obvious with some basic understanding of what the system is doing.

6. A Mediterranean holiday complex has installed a solar powered system to heat water and produce electricity. Answers in jotter.

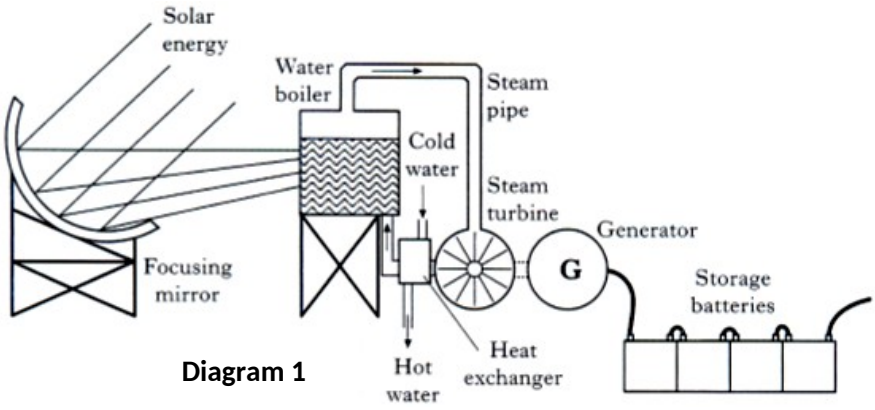


Diagram 1

A) Study the system above and describe all the energy changes taking place in the system. You will be provided with the image (diagram 1).

B) Identify an energy loss within the system above and describe how the system could be made more efficient.

7. Some systems do not convert energy from one form into another. These systems TRANSFER the same form of energy from one place to another. Try to think of some systems that do this and draw universal system diagram for them. Answer in jotter.

What is technology?

Technology influences all our lives so much that there is hardly a thing that is not affected by it. We take the term 'technology' to mean the things or products made by our society.

In our modern world there are so many new products being created that it is impossible for us to keep up and know everything about these developments. As technologists we need some help to use and understand all this technology. The strategy we use is called the *top-down* approach.

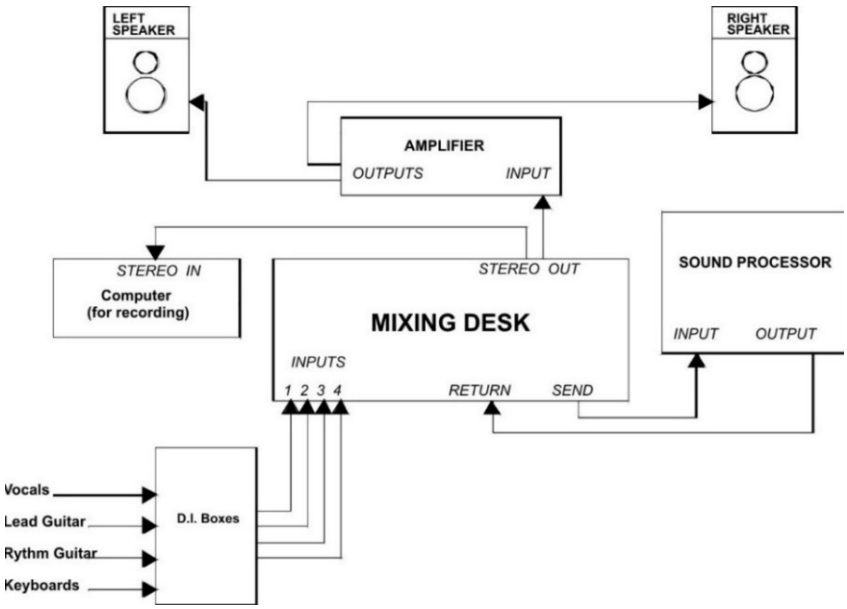


Here we limit the study of technology to what we need to know. For example, to use a toaster all we need to know is that if we plug it in, set the temperature, switch it on and insert a slice of bread then, after a few minutes, we will get toast. However, an engineer designing a new toaster will need to know a lot more and will have an understanding of all the parts that go to make up the toaster and how they relate to one another.

To simplify the difficulties of dealing with complex devices and machines, a method of analysis called the *systems approach* has been developed. It is based on the concept that any technological design or device can be described as a *system* and that this system can be split into smaller parts or *sub-systems*. Each sub-system deals with the *function* of a particular part of the design or device.

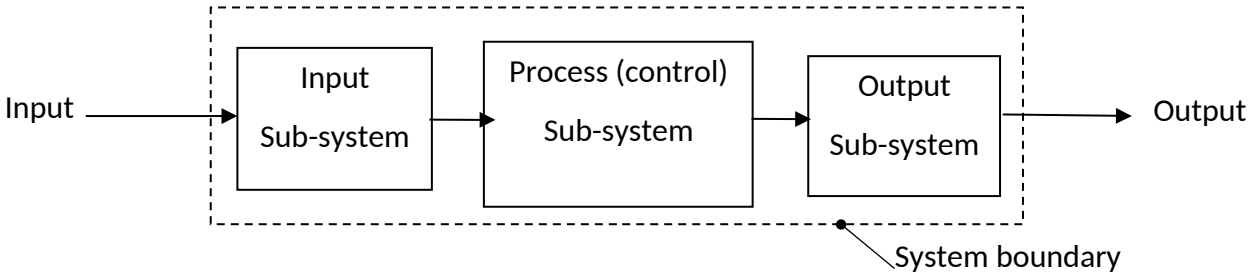
For example, the system diagram opposite describes a sound-mixing system used in the music department of a secondary school.

Even people with no training in the systems approach can look at a system diagram and gain an understanding of the function of the various elements of a system.



Sub-systems

We have seen that the top-down approach is used when studying technology. This deals with technology on a need-to-know basis. For most people, knowing what they must do to make the technology work (input) and what it does (output) is enough. However, we will be using technology to solve problems and so we need to break it down or analyse it in greater depth. To do this, we use a *sub-system* diagram as shown below.



The sub-system diagram shows the *internal* detail of the system. Each box, called a *sub-system*, can be thought of as a system within a system and has its own input and output. The dashed line around the sub-system is called a *system boundary* and this marks the area of interest to us. The 'real world' input and output are shown as arrows entering and leaving the sub-system diagram.

Consider a complicated system of a washing machine. Most systems cannot operate entirely independently of other systems around them. A washing machine has a control system, which is electronic. It also has a water system using valves and pumps, which is a different system to the control system. However, for the washing machine to work, the control system and the water system must overlap each other. This can lead to confusion when drawing system diagrams. In order to isolate the system being considered, a *system boundary* is sometimes drawn. The system boundary is drawn as a dashed line right around the part of the system being considered, thus defining the limits of the system, as shown in the example below.

We would first complete a system diagram showing the inputs and outputs:



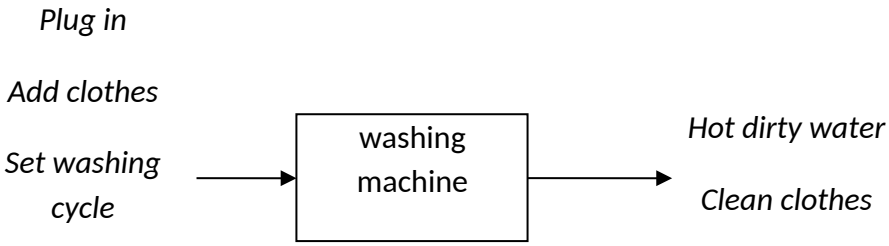
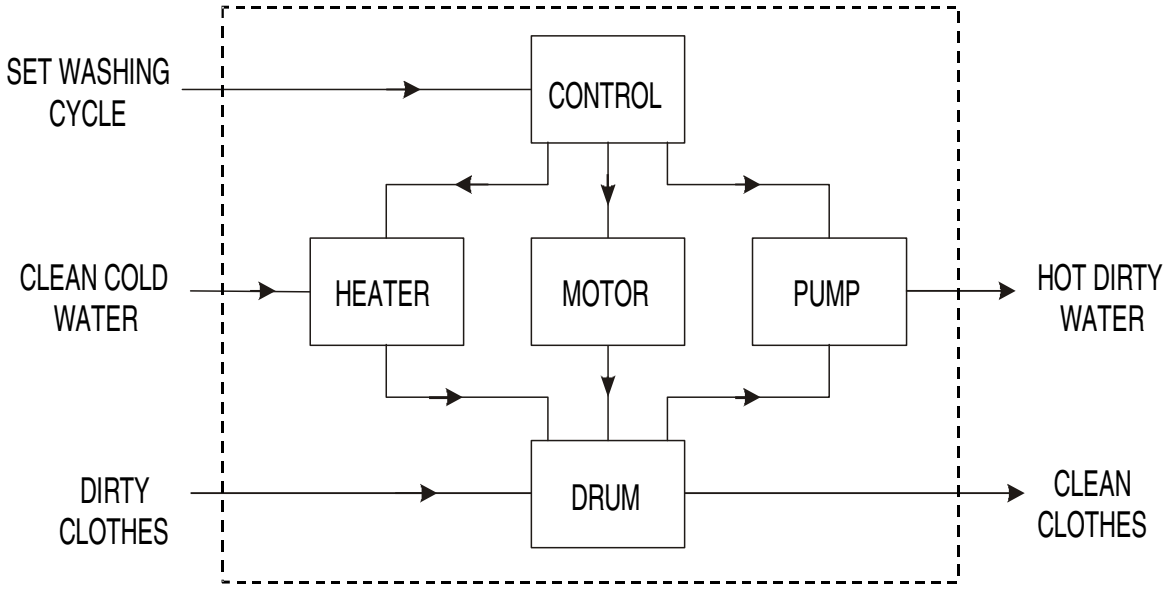
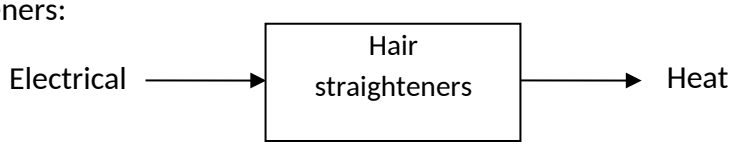


*Diagram above is a **Universal** system diagram*

As we require more detail about the system, it must now be broken down into its sub-systems. This is shown on the simplified sub-system diagram below. Remember a system boundary is a dashed line: - - - - - -

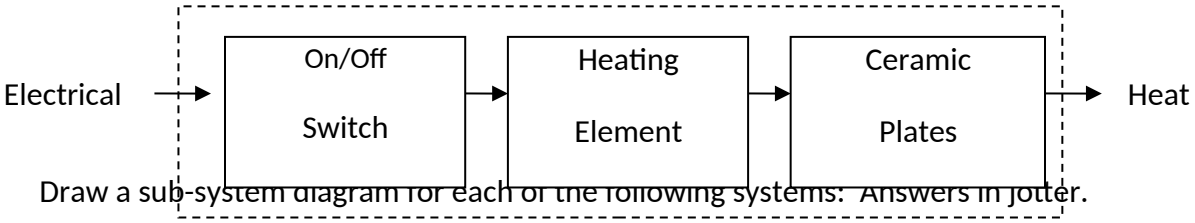


Another example: By using a sub-system diagram it shows more detail of how a system works, this time for a set of hair straighteners:



By creating a universal diagram not much information can be gathered on how the system actually works.

By breaking it down into a **sub-system diagram**, it is clearer how the technology works.



8. Draw a sub-system diagram for each of the following systems: Answers in jotter.

- Vacuum cleaner
- Kettle
- Hand torch (not a bike light)
- Toaster
- Mobile phone (extension task)
- Cash machine (extension task)
- Computer (extension task)



9. Draw a universal system diagram for a house alarm with a door sensor, panic button, keypad arm, control unit, siren and strobe-light alarm. Answer in jotter.

Control Systems

For any system to be effective, it must be adequately controlled. Imagine the problems that might arise in your school if there was no-one:

- in charge.
- taking responsibility for day-to-day activities.
- ensuring that teachers were present and teaching their classes.
- checking that pupils actually went to classes.

Such a scenario could hardly be called a *system*, much less than an education system.

All types of system require some form of control to make the system work properly. A system may be based mainly around people (as in an education system) and is then known as a *soft system*. Systems based on manufactured components (as in a central heating system) are called *hard systems*. The emphasis of the Engineering Science course is on hard systems.

In many cases people do the controlling, especially in soft systems. However, hard systems are usually controlled automatically. Automatic control has become more and more common over recent years as technology has progressed. This type of control releases people from many of the boring and mundane jobs of watching machines operate, with only occasional adjustments to settings being necessary.

Control is absolutely central to the effective functioning of our society, from the streetlight that switches on automatically at night, through the air-conditioning in shops and offices and the autopilot in aeroplanes, to international electronic banking, internet and WiFi enabled appliances.

For this reason, control is also at the heart of the National 4 & 5 Engineering Science course and permeates every aspect of it.

Most hard systems have within them an element that is dedicated to the control of the main part of the system. This control element may be small in relation to the whole system or it may be the main part of the system. In many cases, the control element of a system is important enough to be regarded as a system in its

own right. Therefore, it is called a *control system*. Control systems can be split into two distinct types, either of which may be controlled *manually* or *automatically*.

- Manual control is performed by the actions of humans.
- Automatic control is performed by technological devices (often electronic).

In the Engineering Science course, we are interested in automatic control.

As part of the systems analysis we also normally consider the type of control the system provides. There are two types of control: open - loop control and closed - loop control.

Open-loop control

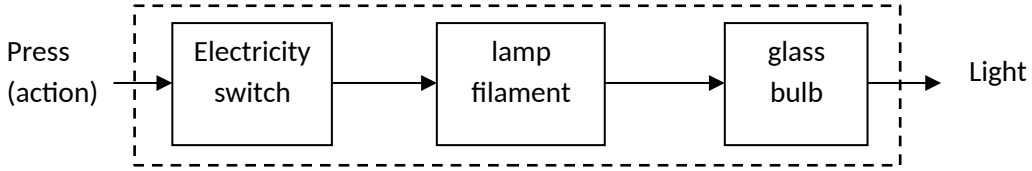
At the simplest level a control system can process an input condition to produce a specified output. This is the simplest acceptable level of control. It is also the most common form of control system, used widely in domestic and industrial systems because it is cheap to install and simple to operate.

In open-loop control the *input* action causes a resulting *output*.

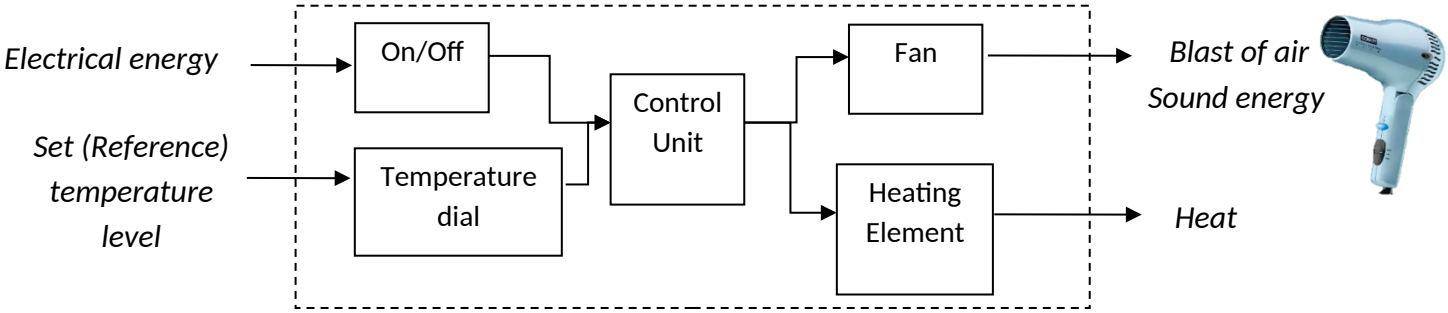
Domestic lighting systems usually have open-loop control. The input is the action of pressing the light switch and the output is light from the filament of the bulb.



A system diagram for the light is shown below. Here it describes a manual *open-loop* control system.



Another good example of this type of control is a hairdryer. The input is the action of pressing the button and the output is the hot air.



Hair dryer system diagram explained: Here the input signal from the on/off and temperature switch is processed to produce the output. The output air is not monitored or adjusted in any way and it is just blown out at whatever temperature the heater warms it to. An open-loop control system is the simplest and cheapest form of control. However, although open-loop control has many uses, its basic weakness lies in its inability to adjust the output to suit the requirements.

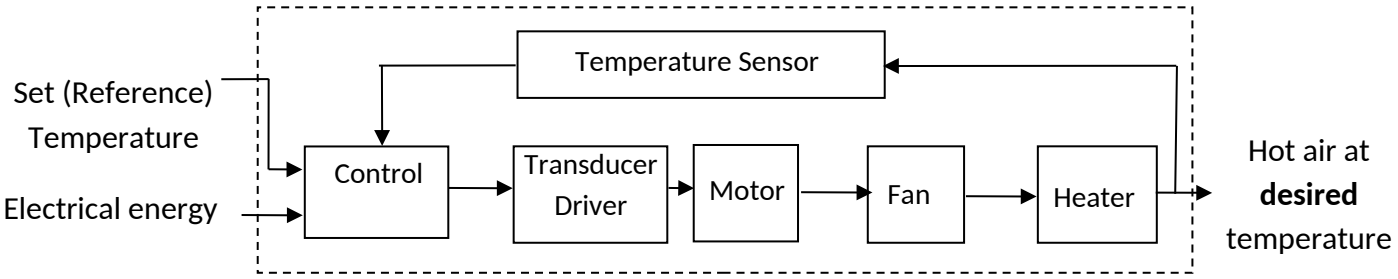
Closed-loop control

This is the most sophisticated form of control. In *closed-loop* control, *the value* of the *output* is constantly monitored as the system operates and this value is *compared* with the set (or reference) value. If there is any *difference* between the actual value and the set value (an *error*), then the *input* to the system is varied in order to *reduce* the *output error* to zero.

Closed-loop control is a more accurate system of control and at the same time more expensive. It employs *self-monitoring*, where a sensor is used to read the condition being controlled and adjust the output if necessary. This monitoring takes place through a *feedback loop*. Here an input sensor checks the output and adjusts it when it does not meet the requirements.

Closed-loop control systems are therefore capable of making decisions and adjusting their performance to suit changing output conditions. An example is a thermostatically controlled fan heater.

The *closed-loop* sub-systems diagram for the heater is shown below.



Note: The Transducer Driver (sometimes referred to as just a driver) provides the output transducers enough power to operate.

In this example a thermostat monitors the output temperature and switches the heater on when it is too cold and off when the temperature is at the required or set level.

System control overview

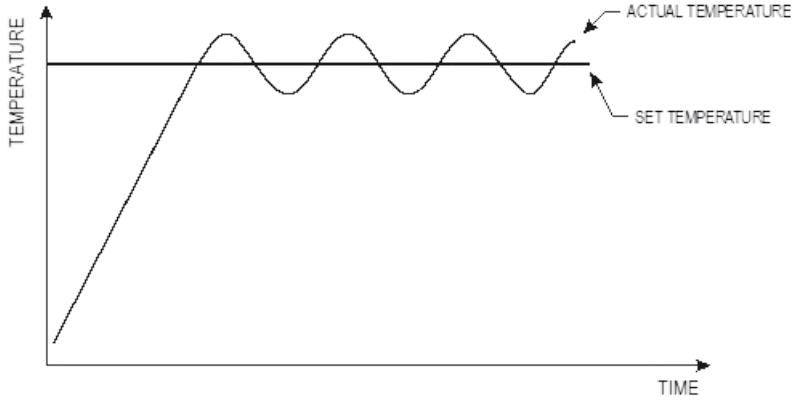
All closed-loop control systems include a sensing sub-system that feedbacks information to the control sub-system. The control sub-system will process this *'feedback signal'* and make a 'decision' on whether to alter the output.

In the system diagram, you should note that the diagram above now forms a continuous loop that can be followed round repeatedly as the system operates. This is why the system is called *closed-loop* and can always be identified by the presence of a feedback loop and the comparison with *open-loop* becomes much clearer as an *open-loop* system **never** has a feedback loop.

Note: In '*closed-loop*' systems, the feedback loop is *not* necessarily a physical link between the sensor and the output. Instead the sensor monitors the *environment* that the output is controlling.

Negative and Positive feedback

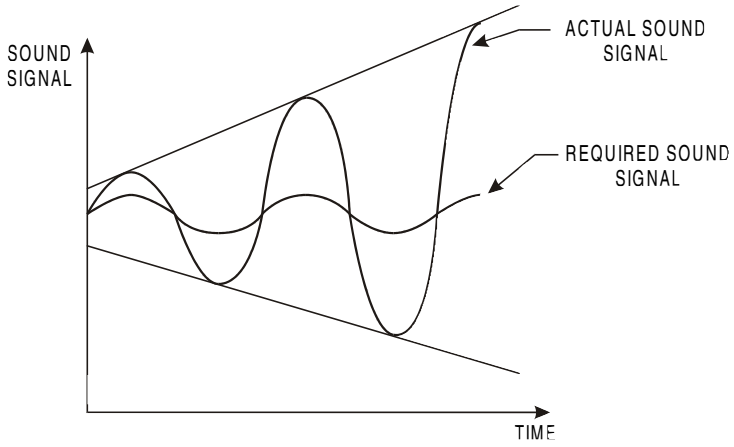
The purpose of closed-loop control is to ensure that the output is maintained at, or as closely as possible to, the desired level. In the case of a central heating system, a graph of the temperature in a room might appear as below:



As can be seen from the graph, the control system is constantly trying to pull the temperature of the room back towards the set temperature level by reducing the error. This type of control uses *negative feedback* to r*educe* the error.

Reinforcing or increasing the error can create the opposite effect, for example in a public-address system when the microphone is held too close to the speakers.

A sound is picked up by the microphone, amplified and then output through the speaker. The amplified sound is then picked up, re-amplified and so on. The net result is a high-pitched sound caused by the feedback and this can be represented by the graph below:

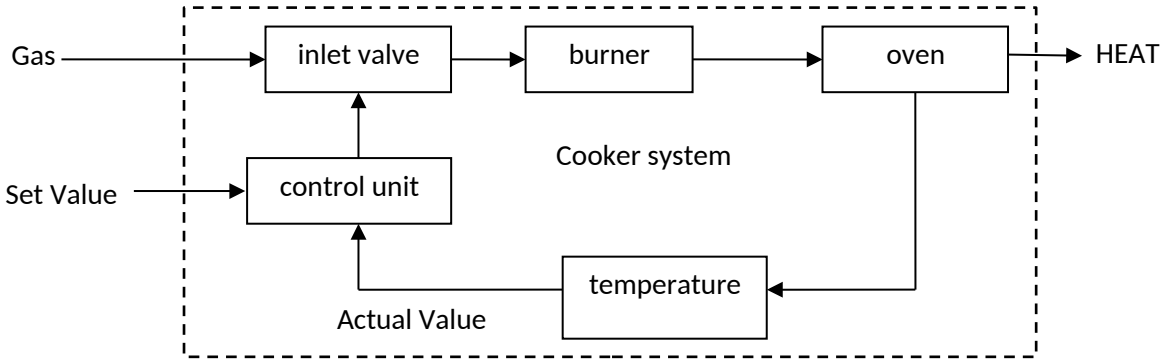


This is an example *of positive feedback* where the error is *increased*. Although positive feedback does have some useful applications, negative feedback is far more widely used in control systems.

The purpose of closed-loop control is to ensure that the output is maintained at, or as closely as possible to, the desired level. For example, the temperature is set on a dial on the control panel of a cooker will be a closed loop control.

A temperature sensor constantly monitors the output (oven temperature) and produces a signal representing the *actual* temperature.

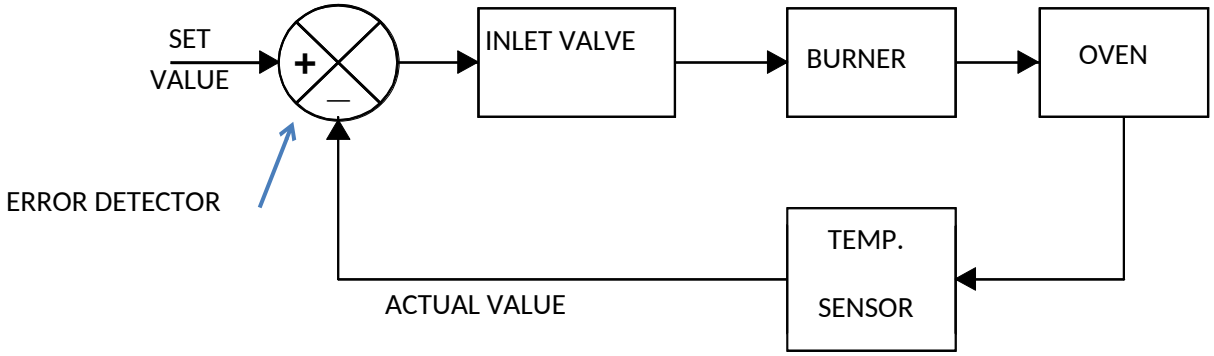
The *actual* signal is fed back to a control element, which compares it with the *set* signal. Any difference between *set* and *actual* signals produces an *error signal*. This causes the control element to either decrease or increase the input (gas flow) in order to reduce the temperature error to zero.



The control unit shown in the system diagram above performs two functions.

- The set and actual values are compared.
- Any resulting error is used to vary the input sub-system as required.

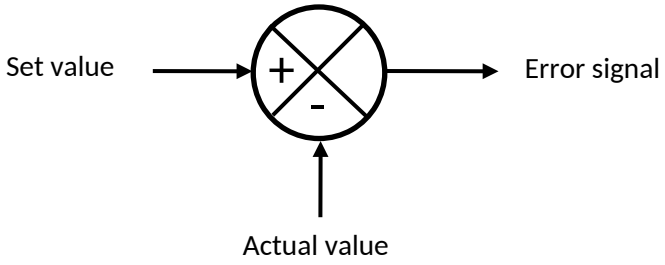
A more detailed diagram, sometimes called a control diagram, may be drawn. This shows the control unit in more detail.



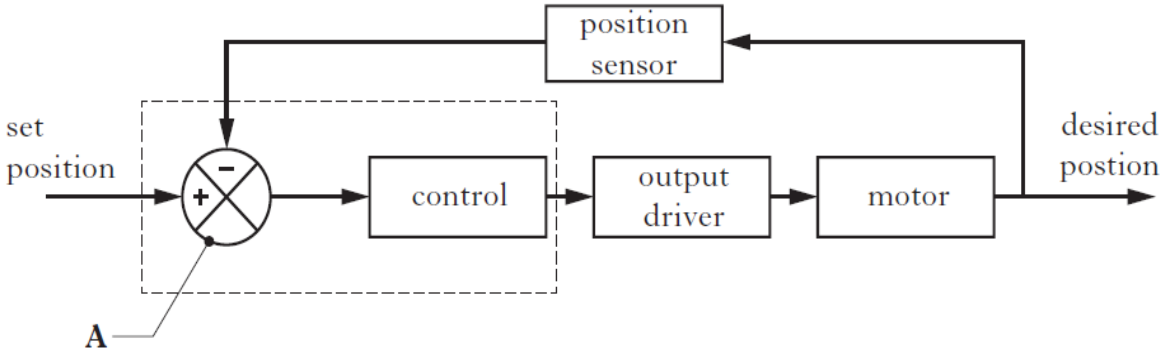
Inside the error detector the actual value is subtracted from the set value and the resultant value is the error signal. The error signal changes the state of the input device in a way that tends to reduce the error.

Error detector diagram

The error detector is a circle divided into 4 parts. Within two of the quadrants, there will be a 'add' and 'minus' graphic. The '+' symbol is the pre-programmed required value and it is always at the 9 o'clock position. The '-' symbol is the actual present value being monitored.

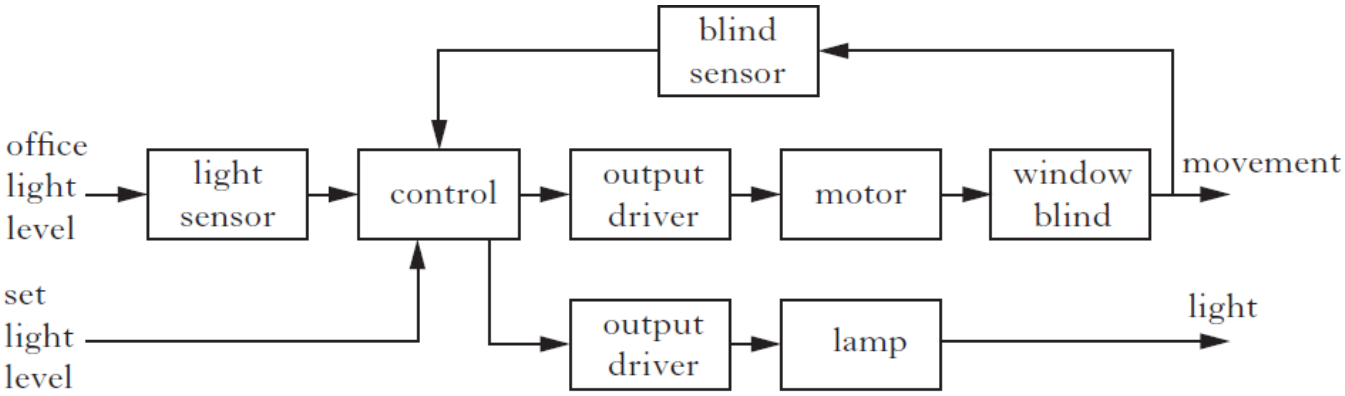


10. A control diagram for the automatic positioning of a telescope is shown. Answer in jotter.



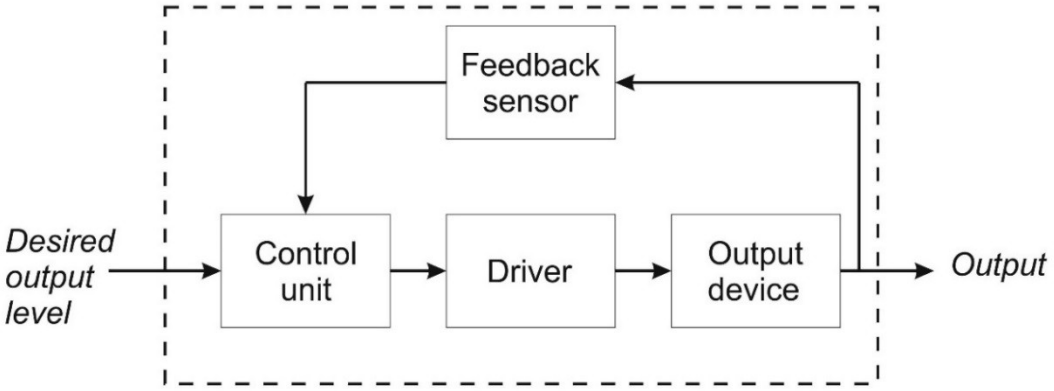
a) State the name of part **A**,
b) State the type of feedback used in the telescope system.
c) Describe why this feedback is used.
d) Describe, with reference to the diagram the operation of the telescope system.

11. The diagram below shows a simplified sub-system diagram for the control of the light level in an office. A light will turn on if the office is too dark or a window blind will close if it is too bright. Answers in jotter.



a) Complete the sub-system diagram shown by adding the system boundary.
b) Describe why a system boundary should be included in a sub-system diagram.
c) Describe, with reference to the diagram, the operation of the window blind.
d) The system uses both open and closed loop control. State the difference between open and closed loop control.
e) State which of the output transducers is operated using closed loop control.

When producing sub-system diagrams for engineering systems in Engineering Science they will usually be in the following format.



The *control unit* will usually be an electronic circuit of some kind. The *driver* will be a device that will amplify the small signal from the control unit so that it is powerful enough to drive the output. The *output device* will usually be either a motor, lamp, or a heater. The *feedback sensor* will usually be either a light sensor, temperature sensor, or a speed sensor.

12. Draw a sub-systems diagram for a fridge with a thermostat, control unit, coolant pump, door switch and light. Answer in jotter.

13. State three examples of open-loop control. Draw a system diagram for each one and show the system boundary. For the third of your diagrams, explain how the system works, using appropriate terminology. Answer in jotter.

14. State three examples of closed-loop control. Draw a system diagram for each one and how the system boundary. For the third of your diagrams, explain how the system works, using appropriate terminology. Answer in jotter.

15. Draw a sub-systems diagram for a cruise control system in a car that will maintain a constant speed even when the car is going up and down hills. Answer in jotter.

16. Draw a sub-systems diagram for a laser cutter that will move a laser to an exact position. Answer in jotter.

17. Draw a sub-systems diagram for a drone that will keep it hovering at a height set by the operator. Answer in jotter.

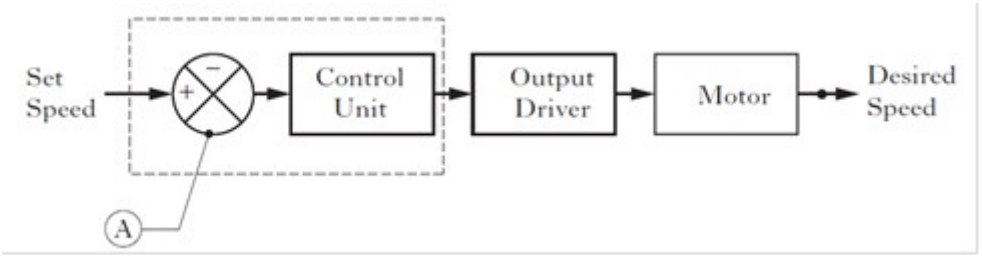


18. The systems approach is often used to help understand engineering problems.
(*a*) Draw the Universal Systems diagram.

(*b*) Draw a system diagram for a toaster. Show the main energy input and the main energy output.

19. A treadmill uses closed loop control to maintain a constant speed.

(a) Describe closed loop control.



An incomplete control diagram for the treadmill is shown above.

(*b*) (i) State the name of symbol A

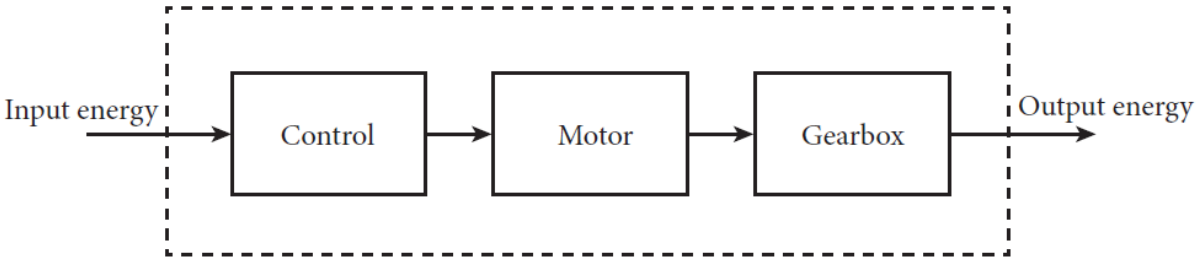
(ii) Complete the control diagram above.

Exam past paper questions. Answers in jotter.

1. A rotating scoreboard is shown.



The main parts of a rotating scoreboard are shown in the diagram below.



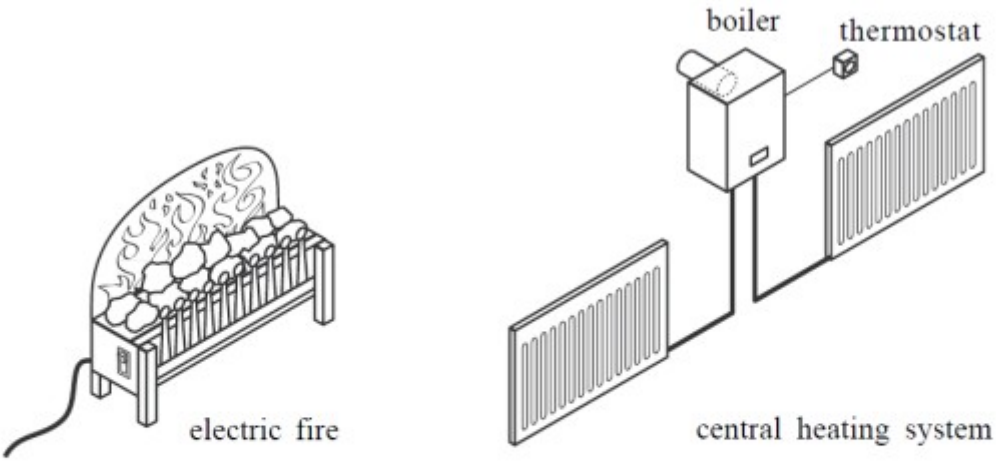
A treadmill uses closed loop control to maintain a constant speed.



(*a*) Describe closed loop control.

2.

3. Two types of heating systems are shown below.



Systems can be placed into two groups, **open loop** or **closed loop** control.

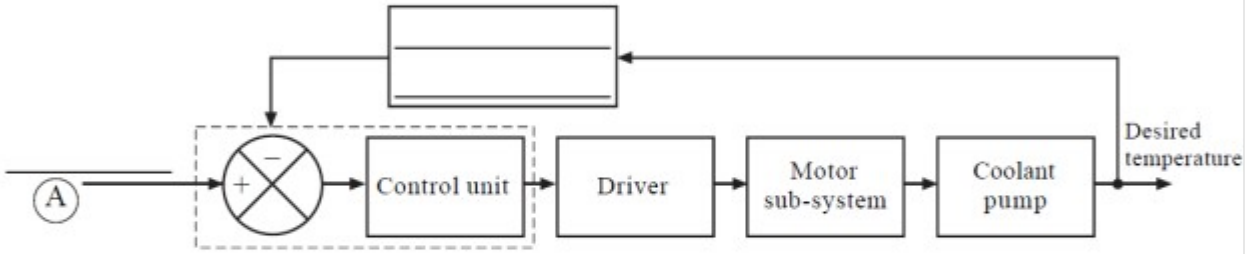
(*a*) State which group each heating system belongs to.

(i) Electric fire _____

(ii) Central heating system _____

2. An incomplete control diagram for a refrigeration system is shown below.





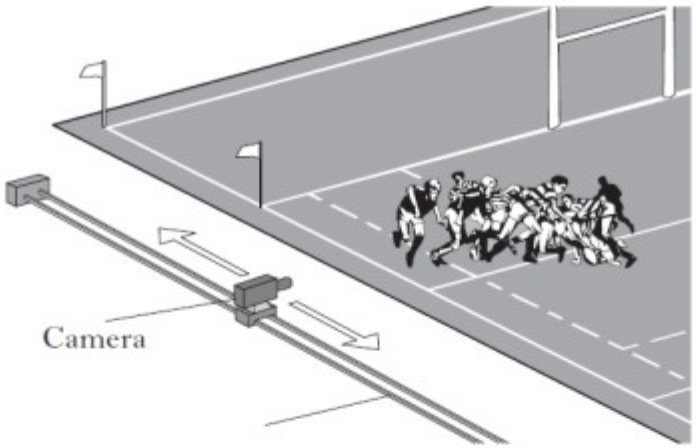
(*a*) Complete the control diagram above.

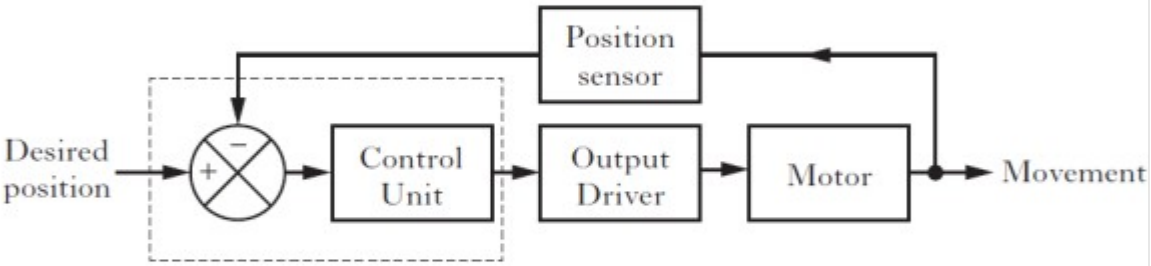
(*b*) Using the correct words from the list below, complete the description of the refrigeration system.

feedback loop *error detector* *compared* *system boundary* *added* *inputted*

Input (A) is _____ with the actual temperature and the _____ decides whether or not to turn the motor on and cool the system.

5. A mobile pitch-side camera system is used during a rugby match.





(*a*) Describe, with reference to the control diagram, the operation of the system.

The camera system makes use of closed loop control.

(*b*) Explain the difference between open loop and closed loop systems.

Student calculation worksheets. Answers in jotter.

Assignments: what is energy?

1. Why do humans and machines need energy?
2. What happens to the energy in a system if it is never destroyed?
3. What is the ultimate source of energy for the Earth?
4. Describe briefly the three main ways that the sun's energy is available on Earth.
5. Give some examples of energy sources that can be used from each of the three groups.

Assignments: fossil fuels

1. What are coal, oil and natural gas made from and why are they called fossil fuels?
2. What is the danger to the Earth's energy supply if we continue to burn coal, oil and gas?
3. What other resource apart from energy can be extracted from oil?

Assignments – nuclear energy

1. What is the name of the material that is used in a reactor to generate nuclear power?
2. What are the problems associated with the waste materials produced by a nuclear reactor?

Assignments – energy and the environment

1. What is 'global warming'?
2. Describe the way that the greenhouse effect is warming the Earth.
3. How is acid rain produced?
4. Why does acid rain have an effect on animals as well as plants?
5. What must be done to reduce the effects of acid rain and global warming?

Assignments: renewable energy sources

1. What is the difference between a finite and an infinite energy source?
2. Why are fossil fuels thought of as non-renewable, or finite, energy sources?
3. What is a renewable energy source?
4. Name three renewable energy sources.
5. Name a machine that has been used in the Netherlands for many years that harnesses the energy of wind.

Assignments – solar energy

1. Describe three ways in which it is possible to harness solar power.
2. If a word has 'photo' (for example photosynthesis) at the start of it, what does 'photo' refer to?
3. Where would be a good location for a solar power generator

Assignments – wind energy

1. Name one method of transport that harnesses the wind's energy.
2. How can wind farms cause environmental problems?
3. Describe the way that electrical energy is generated using a wind turbine like the one shown above.

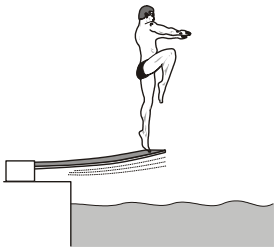
Assignments – hydroelectric energy

1. Describe the way that electricity is generated in a hydroelectric power station.
2. What is a possible disadvantage of hydroelectric schemes?

Assignments – energy from the sea

1. Describe two ways that the sea's energy can be harnessed.
2. Describe one negative aspect of harnessing tidal energy.

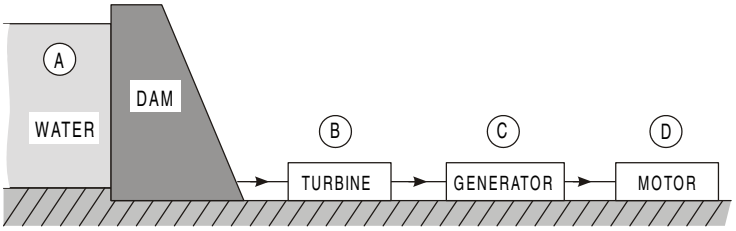
Assignments: identifying different forms of energy



1. When an Olympic diver stands on a diving board 10 m above the pool, what form of energy does he possess?
2. When the diver jumps on the diving board it will bend. What form of energy does the board now possess?
3. When the diver is moving upwards, what form of energy does he now possess?
4. When the diver is at the highest point of his dive why does he not have any kinetic energy?
5. A heating element is used to make sure that the water temperature in the pool is maintained at a suitable level. Copy and complete the simple systems diagram below, showing one energy input and one energy output for the heating system that will solve this problem.



6. Name four sources of energy that can be converted (directly or indirectly) into electrical energy.
7. The diagram below shows a representation of a hydroelectric power station.

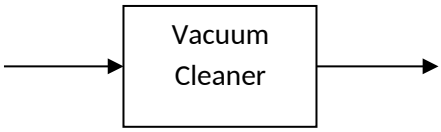


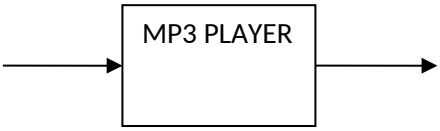
Name the form of energy at each of the stages A, B, C and D.

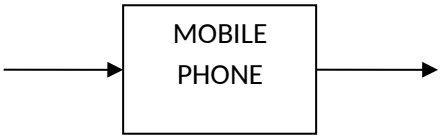
Student homework 1

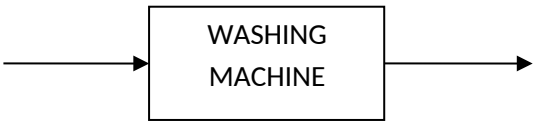
Draw a system diagram showing the inputs to and the outputs from each of the following.





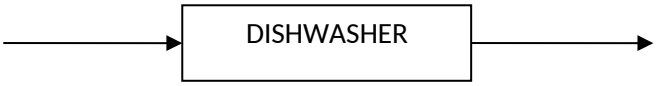


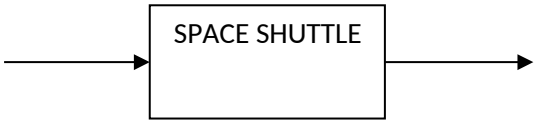














Student homework 2

Complete the following system diagrams.

